

Lecture 4 Part 2:

Considerations for Data

Collection & Data

Analysis

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Part 2's Lecture Contents

- Data types (terms / definitions) to be aware of
- Data collection preparation & management
- Data visualisation (to aid analysis)
- The generalisability of results

Experimental Data

Experimental Data

- Experiments **generate data** in a systematic and controlled manner
- Experimental data is used to **accept** or **reject** a hypothesis:
 - Accept if the data supports it, reject if it does not support it

Types of Experimental Data

Quantitative data: that expresses a numerical quantity

Examples: compilation time, classification accuracy (%), number of errors, reaction time, task success (Y/N), etc.

Qualitative data: that is non-numerical

Examples: interview transcripts, observation notes, ethnographic diaries, documents, videos, etc.

Experiments should be designed with data in mind

Common mistake: designing experiments that do not yield useful data

For example: measuring the wrong thing, measuring time in seconds instead of milliseconds.

Common mistake: gathering data that doesn't fit the analysis approach

For example: complex experiment designs with too many IVs so it becomes difficult to isolate the effects of each factor

Types of Data

Types of data — Continuous / Discrete data

- **Continuous data:** a set of numerical data where the values can be anywhere in a range, including fractional values
 - Examples include, time taken, height, etc
- **Discrete data:** A set of numerical data where the values are at specific points rather than anywhere along a range
 - For instance, a user can have 12 or 13 errors but never 12.5

Types of data — Nominal data

Categorical / Nominal data: a group of objects or ideas that can be collectively grouped on the basis of a particular characteristic

- E.g. gender, hair colour, eye colour, etc

A nominal group only has members and non-members: nothing more can be said about the members other than they are part of the group

- Nominal categories cannot be numerically organised or ranked

Types of data — Ordinal data

- **Ordinal data:** classifies variables into categories which have a natural order / rank
 - E.g. satisfaction on a scale of “very unsatisfied, unsatisfied, neither unsatisfied nor satisfied, satisfied, very satisfied”

Ordinal data is distinguished from the nominal scale by having a ranking

Types of data — Interval data

- **Interval data:** measured along a numerical scale that has equal intervals between adjacent values
 - E.g. temperature measured in celsius
- Interval data shows you the order/direction and difference between values
- There is no true zero in interval data:
 - True zero refers to a scale where 0 indicates the absence of something.
 - Consider temperature — 0 degrees is still a temperature (of heat)

Types of data — Ratio data

Ratio data: measured along a numerical scale that has equal intervals between adjacent values AND has a true zero

E.g. height, weight, time

A Summary

THE FOUR LEVELS OF MEASUREMENT:

	Nominal	Ordinal	Interval	Ratio
Categorizes and labels variables	✓	✓	✓	✓
Ranks categories in order		✓	✓	✓
Has known, equal intervals			✓	✓
Has a true or meaningful zero				✓

Data Collection & Management

We want data collection to be planned and organised

- We don't want our data collection to be random
- We don't want to do our experiment only to find out:
 - We collected the wrong data
 - We didn't collect data to sufficient accuracy levels
 - The collected data format is unusable

While these sound obvious, these are easy mistakes to make

We want data collection to be planned and organised

- We also don't want to collect data only to discover we have broken some data collection protocol at your employment
 - E.g. we need to ensure we collect data ethically

(1) Prepare analysis scripts early

- Prior to running an experiment, we can set up our data analysis scripts
 - Help us to know planned data capture from experiment
- So, before you run the study, if possible, select data analysis methods and setup statistical scripts to analyse the data
 - Generate a set of fake data
 - Use this to create the analysis scripts

This method saved time post-experiment and can highlight shortcomings of your existing approach

(2) Run a pilot test of the study

Beneficial for testing the experiment setup and to test the data collection method

- Can be done with target study participants (early insight into the actual data you might collect / understanding of the experiment)
- Can be done with a knowledgeable colleague (who can critique experimental design, setup, data collection, etc)

Data Visualisation & Exploration

Data visualisation and exploration

Two ways we can consider this:

- Visualisation, and exploration, of our data during the analysis process
- Visualisation of the data when presenting it via report, etc

Both are valuable things to consider

Data visualisation and exploration — during analysis

- We just spoke about preparing analysis scripts before we've collected any actual participant data
 - Can do this when we know the investigated hypothesis in advance

But what if a hypothesis is a limitation?

A Hypothesis is a Liability — Yanai and Lercher

- Discussed “day science” and “night science”
- Day science — the structured testing of predefined hypothesis we have been discussing
- Night science — “we explore the unstructured realm of possible hypotheses, of ideas not yet fully fleshed out”

<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-020-02133-w>

A Hypothesis is a Liability — Yanai and Lercher

- “In day science, we falsify hypotheses and observe which are left standing; in night science, we create them”
- Both have their uses and various advantages / disadvantages
- These are complementary methods — the shortcomings of night science are corrected by day science (and vice versa)

Yanai and Lercher did an experiment to demonstrate this

A Hypothesis is a Liability — Yanai and Lercher

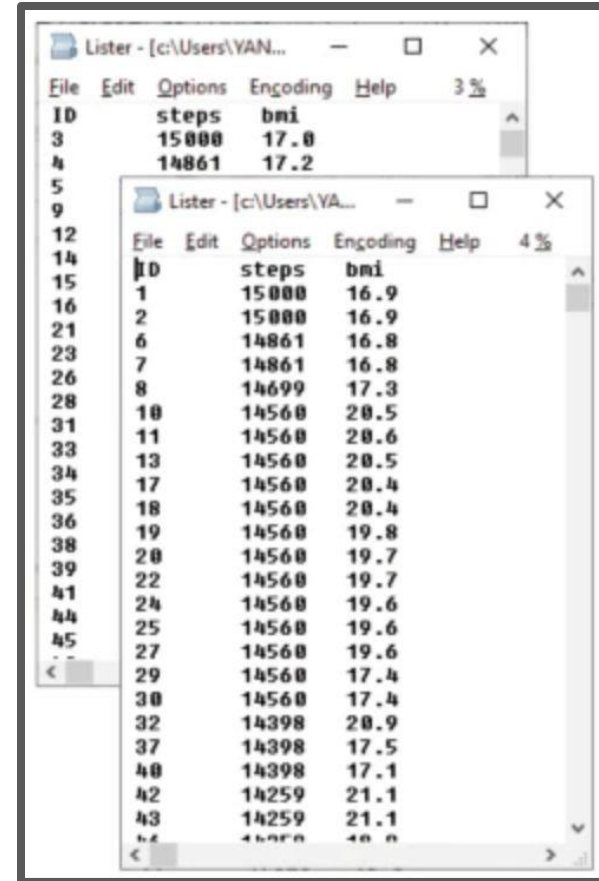
- Wanted to demonstrate that the mental focus on a specific hypothesis prevent us from making a discovery
 - To test this, they made up a dataset and asked students to analyse it

A Hypothesis is a Liability — Yanai and Lercher

They created a dataset containing the body mass index (BMI) of 1786 people, together with the number of steps each of them took on a particular day, in two files:

- One for men
- One for women

Students placed into two groups and given this data



The image shows two overlapping Notepad++ windows displaying CSV data. The top window, titled 'Lister - [c:\Users\YAN...', shows the first three rows of a dataset with columns 'ID', 'steps', and 'bmi'. The bottom window, also titled 'Lister - [c:\Users\YAN...', shows a larger portion of the same dataset, starting from row 10.

ID	steps	bmi
3	15000	17.0
4	14861	17.2
10	15000	16.9
11	15000	16.9
12	14861	16.8
13	14861	16.8
14	14699	17.3
15	14560	20.5
16	14560	20.6
17	14560	20.5
18	14560	20.4
19	14560	20.4
20	14560	19.8
21	14560	19.7
22	14560	19.7
23	14560	19.6
24	14560	19.6
25	14560	19.6
26	14560	19.6
27	14560	17.4
28	14560	17.4
29	14398	20.9
30	14398	17.5
31	14398	17.1
32	14259	21.1
33	14259	21.1

A Hypothesis is a Liability — Yanai and Lercher

- The students in 1st group were asked to consider three specific hypotheses:
 - (i) that there is a statistically significant difference in the average number of steps taken by men and women,
 - (ii) that there is a negative correlation between the number of steps and the BMI for women
 - (iii) that this correlation is positive for men

Also asked if there was anything else they could conclude from the dataset

A Hypothesis is a Liability — Yanai and Lercher

The students in 2nd, “hypothesis-free,” group were simply asked:

- What do you conclude from the dataset?

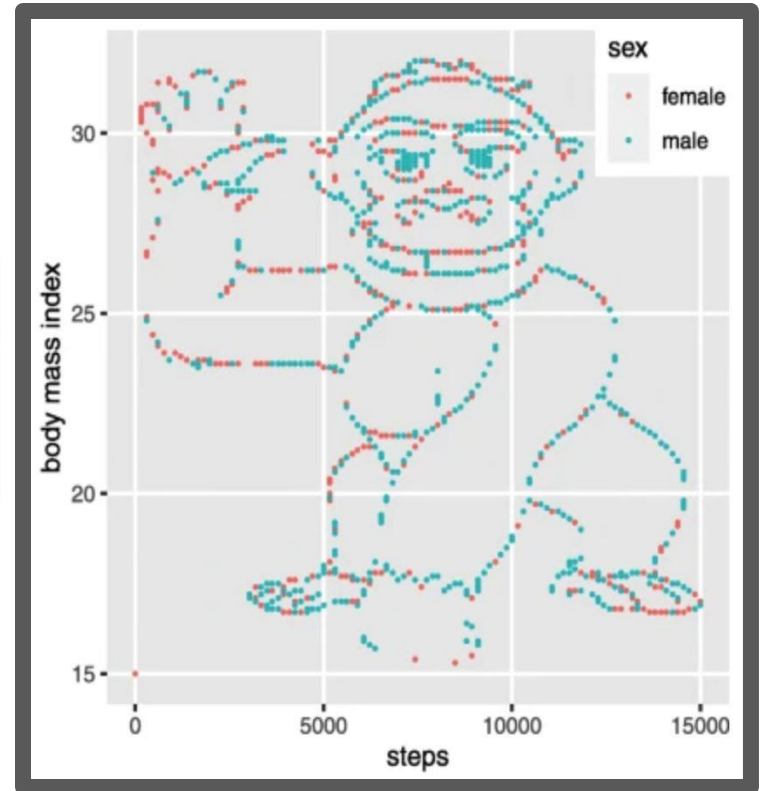
So what happened?

A Hypothesis is a Liability — Yanai and Lercher

If you did the most basic visualisation of the data (plot steps on X, BMI on Y) you got this:

	Gorilla <u>not</u> discovered	Gorilla discovered
Hypothesis-focused	14	5
Hypothesis-free	5	9

Why generalisability and scope of results matters



Important to think about the scope of claims made

A large part of what you do as a researcher is attempt to convince your peers that your results are correct / valid

- An important part of this is ensuring you **do not overclaim** your results

Overclaiming is where you make a claim that you prove/show more than your results actually do

Generalisability of experiments

First: A recap experimental factors

Independent Variables: the factors you deliberately change

Dependent Variables: the data you are collecting

Control Variables: factors that do not change to ensure a “fair” comparison

First: A recap experimental factors

Random Variables: factors you knowingly allow to randomly change/vary

Confounding Factors: factors that change and which impact the results (usually only identified after you have collected the data)

- Factors which could have influenced the relationship between the independent and dependent variable

What do we want an experiment to show?

- A **change** in the values of the **independent variable** will result in a change in the values of the **dependent variable**

We want to be sure that:

- Any **change** in the value of the **dependent variable** is a result of the change in the value of the **independent variable** (rather than a result of change in the value of other factors)

How to ensure changes in IV caused changes in the DV ?

1. Control variables — explicitly set the values of all other potential variables in the experiment, so that they do not affect the DV
2. Post-experiment — do not control, but analyse the data with respect to other factors

Ensuring changes in IV caused changes in the DV

3. Randomisation in the task — attempt to ensure the effects of other variables are equivalent overall “average out” the effects of other potential causes through randomisation

Note: More control means less generalisability of results
- Means our results only are applicable to the fixed control values

Similar to “too specific a hypothesis”

- This is similar in nature to the idea we spoke about last week of having “too specific a hypothesis”
 - The same principle but when it comes to collecting our data or presenting / discussing our results

Example — graph layout algorithms

RQ: “Which graph layout algorithm is best?”

Criteria: ease of following a path between two nodes

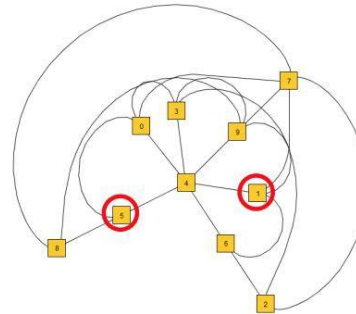
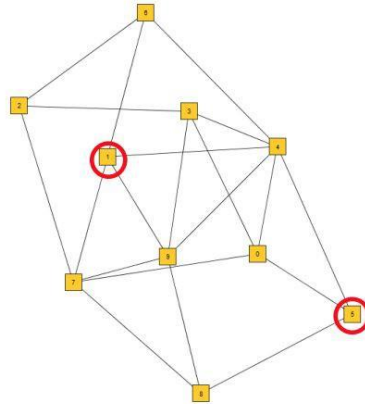
Task: Find the shortest path between two nodes

Independent variable: layout method

Conditions: force layout, radial layout

Dependent variable: time taken to find correct answer

Control variables: the graph, the specified nodes (1 and 5), the node size



Example — graph layout algorithms

Participants have to find the shortest path between these two nodes

RQ: “Which graph layout algorithm is best?”

Criteria: ease of following a path between two nodes

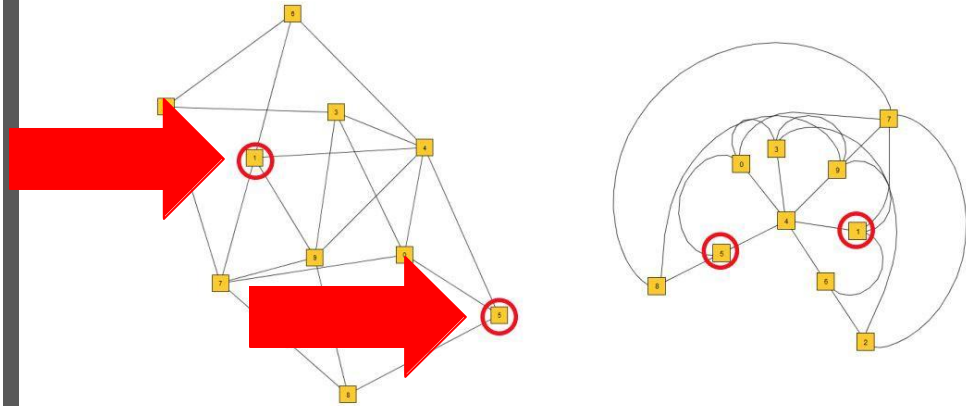
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Example — graph layout algorithms

This is fixed for all participants (our control variable)

RQ: “Which graph layout algorithm is best?”

Criteria: ease of following a path between two nodes

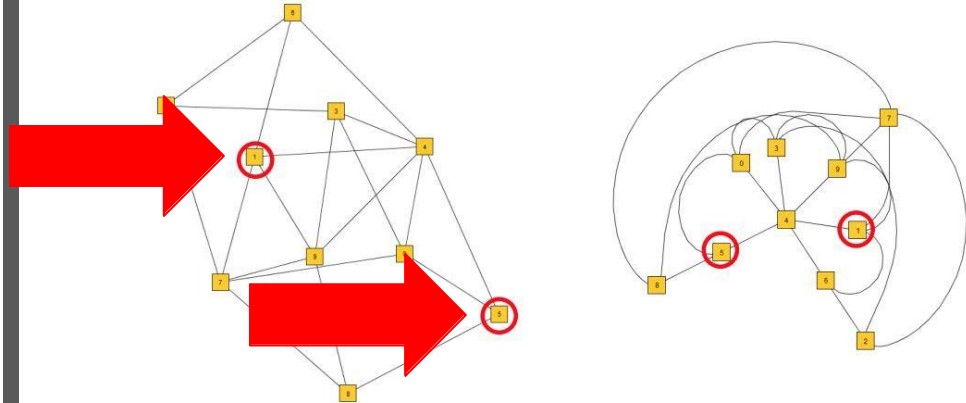
Task: Find the shortest path between two nodes

Independent variable: layout method

Conditions: force layout, radial layout

Dependent variable: time taken to find correct answer

Control variables: the graph, the specified nodes (1 and 5), the node size



Suppose we run this experiment and conclude the following:

“Radial layout is better for a graph of 10 nodes and 20 edges with highest degree 6, for the task of finding the shortest path between two specified nodes (1 and 5)”

Suppose we run this experiment and conclude the following:

“Radial layout is better for a graph of 10 nodes and 20 edges with highest degree 6, for the task of finding the shortest path between two specified nodes (1 and 5)”

- This result isn't very generalisable — we've controlled the experiment too tightly so we have a very context specific result
 - If X and if Y and if Z and if A and if B ...

How could we make this result more generalisable?

(1) Control variables — explicitly set the values of all other potential variables in the experiment, so that they do not affect the DV

We can increase the variability in the control variables so as to make the results more generalizable

- Use more than one graph
- Use more than one task

How could we make this result more generalisable?

(2) Post-experiment — do not control, but analyse the data with respect to other factors

If we think that graph density might have been a confounding factor:

- Analyse the according to density of the graphs – high, medium, low

If we think that gender might have had an effect:

- Analyse data from females separately from male participants

How could we make this result more generalisable?

(3) Randomisation in the task — attempt to ensure the effects of other variables are equivalent overall “average out” the effects of other potential causes through randomisation

We can mitigate against possible confounding factors through randomisation

- Randomly choose different node pairs for each task
- Choose a random orientation for each graph drawing

Any Questions?

Summary of this lecture

- Discussed various key ideas in:
 - Types of data (words to know and be able to use)
 - Data collection and management (as part of the experimental process)
 - Data visualisation and exploration (during analysis)
 - The generalisability of results